

Name: _____ Date: _____

Period: _____

Direction of Motion and Slope Check — Key

A particle moves with $f(t) = (2t - t^2, t^2 - 4t + 3)$ for $0 \leq t \leq 4$.

1. At $t = 0.5$, is the particle moving right or left? Up or down?

$$x(0) = 0 \quad x(1) = 2 - 1 = 1 \quad x(t) \text{ is } \textbf{increasing} \text{ near } t = 0.5 \Rightarrow \text{moving } \textbf{right}$$

$$y(0) = 3 \quad y(1) = 1 - 4 + 3 = 0 \quad y(t) \text{ is } \textbf{decreasing} \text{ near } t = 0.5 \Rightarrow \text{moving } \textbf{down}$$

2. ARC of $x(t)$ and $y(t)$ on $[0, 3]$; slope of curve.

$$x(0) = 0 \quad x(3) = 6 - 9 = -3 \quad \Delta x = -3$$

$$y(0) = 3 \quad y(3) = 9 - 12 + 3 = 0 \quad \Delta y = -3$$

$$\text{Slope of curve} = \frac{\Delta y}{\Delta x} = \frac{-3}{-3} = 1$$

Teacher Note

Common error: students compute $\Delta y/\Delta t$ or $\Delta x/\Delta t$ and report one of those as “slope.” Redirect: “Slope of the curve uses the ratio of the change in the y -coordinate to the change in the x -coordinate, not the change in t .” Note that the slope here is 1 even though both Δx and Δy are -3 ; students should verify that $(-3)/(-3) = 1$, not -3 .